



THE
SANDBOX
TECHNOLOGY CLUB

The Sandbox

An afterschool club for students who want to build things.

Proposal — Academic Year 2025/26

Students come in, work on their own projects, and leave having built something real. No fixed curriculum. No pressure. Just a space, a guide, and the freedom to create.

What this is

The Sandbox is a weekly afterschool club where students build their own technology projects at their own pace. It is open to everyone from KS3 to A Level, regardless of prior experience. A newcomer who has never written a line of code is as welcome as an A Level student building a full website.

The guiding principle is simple: students own their projects. The teacher is there as a guide, not an instructor. There is no set curriculum, no homework, and no grading. What there is, is a room where things get made.

What makes it different

Most coding clubs follow a taught programme. Everyone does the same thing at the same time. Students who are ahead get bored. Students who are behind get lost.

The Sandbox works differently. Each student has their own project. Sessions follow a consistent rhythm, but every student is on their own trajectory. A student who wants to build a game builds a game. One who wants to prepare for a certification works on that. One who has no idea what they want to build gets helped to find it.

The teacher circulates, asks questions, and nudges. The session structure creates accountability without pressure.

Who it is for

Open to all year groups. No selection. No prior experience required. Four levels of student are expected:

Newcomer

Never coded before. Goal for session one: one working program. That is it.

Beginner

Can write basic code. Gets stuck when things break. Needs examples to modify and a clear scope.

Intermediate

Comfortable with Python or JavaScript. Googles errors. Needs architecture guidance and someone to think with.

Advanced

Designs systems, not just scripts. Debugs independently. Needs ambitious scope and a sparring partner.

How a session works

Every session is two hours and follows the same rhythm. The structure creates momentum without being rigid.

Time	Moment	What happens
0 – 10	Stand-up circle	Each student says what they are building, what they did last week, and what they will do today.
10 – 80	Build time	Students work independently. The teacher circulates maximum 5 minutes per student. Rule: never touch their keyboard.
80 – 95	Spark session	For any student without direction: open the project gallery together and ask one question which of these made you want to click on it?

95 – 105	Commit check	Each student writes one sentence: by next session I will have X. One demonstrable thing. No vague goals.
105 – 120	Show and tell	One student per week shares their screen working or broken. Questions from the group only. The teacher closes with one specific observation.

What students can build

Projects are entirely student-chosen. The club maintains a curated Project Spark Gallery, a browsable collection of real project ideas across six categories, four levels, and multiple languages, which the teacher uses when a student needs inspiration.

Project categories

- Web (dashboards, portfolio sites, tools)
- Games (pygame, canvas, logic games)
- Data (visualisations, personal trackers)
- AI (chatbots, image classifiers)
- Hardware (Arduino, Raspberry Pi)
- Tools (productivity apps, utilities)

Academic pathway

- Python Institute Certification (PCEP / PCAP)
- JavaScript Certification — OpenEDG JS Institute
- Object-Oriented Programming in Python
- Algorithms in Python
- Luxembourg Informatics Olympiad (OIL)

How students get unstuck

Before a student can ask the teacher for help, they work through four layers in order. AI sits at Layer 3 — a professional tool, not a first resort.

Layer	Source	How to use it
1	Yourself	Read the error. Google it. Try one fix. Spend at least 10 minutes before moving on.
2	A peer	Describe the problem out loud to another student. Most bugs get solved in this step.
3	AI tools	Claude, ChatGPT, Copilot. Paste the error and your code. Read and understand the answer before applying it. Rule: you may use AI on any project — if you can explain every line of the output.
4	The teacher	Now we talk. First question: what did layers 1, 2 and 3 tell you?

This is not gatekeeping. It is how developers actually work - and learning to use AI well is part of the skill set the club builds.

What students get out of it

Tangible

- Real projects in their portfolio
- Internationally recognised certifications
- Olympiad participation and results
- Something to talk about at university interviews

Intangible

- Confidence from shipping something that works
- Independent problem-solving instinct
- The habit of finishing what they start
- A peer group that builds things

What the school gets

- A stronger STEM identity with concrete student output to showcase
- National competition participation through the Luxembourg Informatics Olympiad
- An annual showcase event where students present finished projects
- Cross-curricular opportunities: visualisation projects with Arts, sound projects with Music
- University readiness infrastructure that goes beyond classroom assessment

Logistics

Frequency	Weekly, one session per week
Duration	2 hours per session
Location	Designated classroom, consistent week to week
Membership	Open to all KS3, IGCSE, A Level. Interest and commitment only.
Hardware	School-provided computers. Students may bring own devices (Arduino, Pi, etc.)
Software	Free tools: Python, VS Code, Trinket.io, browser dev tools
Cost to school	None beyond the room and the teacher's time
Annual event	End-of-year showcase: students present projects to school community

What is being asked for

This proposal asks for three things:

- Permission to run the club, so it can be advertised to students.
- A room: the same classroom each week, with working computers and reliable internet.
- Nothing else: no budget, no additional staff, no resources beyond what the school already has.

The club is designed to be zero-cost and self-sustaining. It will grow through word of mouth as students show each other what they have built.

The Sandbox gives students the one thing that no lesson can fully replicate: unstructured time to build something they actually care about, with a guide nearby when they get stuck.

That is enough to change how a student thinks about what they are capable of.